

AKHIL VATS

GenAI Engineer | LLM Applications | RAG | GraphRAG | Agentic AI

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PROFESSIONAL SUMMARY

GenAI practitioner with hands-on, self-directed experience building LLM-powered applications, RAG pipelines, GraphRAG systems, and agentic workflows in Python. Built Digital Vakil, a legal-domain RAG system indexing 32,678 chunks across 30 source documents (18 reference books and 12 Indian legal Acts, including BNS, BNSS, BSA, Constitution of India, Companies Act, Consumer Protection Act, Contract Act, IT Act, Motor Vehicle Act, POSH Act, RTI Act, and Transfer of Property Act), improving retrieval precision from 0.63 to 0.78 (RAGAS) through hybrid retrieval and query optimization.

Experienced with LangChain, LangGraph, Neo4j, ChromaDB, BM25, FastAPI, Docker, and AWS EC2. Brings 5+ years of professional experience in customer operations, quality assurance, SLA management, and team leadership, combined with 19 months of applied AI/ML training and self-directed project work since Feb 2025.

TECHNICAL SKILLS

Programming: Python

Generative AI: LLMs, RAG, GraphRAG, Agentic AI, ReAct, Prompt Engineering, Guardrails

Retrieval & Search: BM25, Vector Search, Semantic Search, Hybrid Retrieval, MMR, Re-ranking, Metadata Filtering, Embeddings, ChromaDB, FAISS

Knowledge Graphs: Neo4j, Knowledge Graphs, Graph Retrieval

LLM Frameworks: LangChain, LangGraph, LlamaIndex, AutoGen, CrewAI, MCP (Model Context Protocol)

LLM APIs: OpenAI API, Azure OpenAI, Gemini API, Anthropic API

Backend & Deployment: FastAPI, REST APIs, Docker, AWS EC2, Streamlit, Redis

Evaluation & LLMops: RAGAS, DeepEval, Retrieval Evaluation, Hallucination Detection, LangSmith, MLflow, GitHub Actions, CI/CD, Observability

Machine Learning: Scikit-learn, XGBoost, PyTorch, Pandas, NumPy, NLP, Deep Learning

Tools: Git, GitHub, n8n, Dify, Cursor AI

TRAINING & SKILL DEVELOPMENT

GenAI & Data Science Training — Learnbay

Feb 2025 – Present | Noida, India

- Completed applied coursework and self-directed projects in Python, Machine Learning, Deep Learning, NLP, and Generative AI.
- Built AI/ML projects involving LLMs, RAG, GraphRAG, Agentic AI, LangChain, LangGraph, vector databases, and API development (see Selected GenAI Projects below).
- Practiced end-to-end AI workflows including data ingestion, retrieval pipeline design, evaluation, debugging, and deployment.

SELECTED GENAI PROJECTS (PERSONAL / SELF-INITIATED)

Digital Vakil — GraphRAG-Powered Legal AI Assistant

Python, LangGraph, LangChain, Neo4j, ChromaDB, BM25, OpenAI, FastAPI, Docker, AWS EC2

- Built an end-to-end legal AI assistant covering 12 Indian legal domains (BNS, BNSS, BSA, Constitution of India, Companies Act, Consumer Protection Act, Contract Act, IT Act, Motor Vehicle Act, POSH Act, RTI Act, Transfer of Property Act), ingesting 30 source PDFs (18 reference books, 12 Acts) into 32,678 indexed chunks.
- Developed a document ingestion pipeline with PDF extraction, cleaning, section-aware chunking, and metadata extraction (Act, section, page) for source traceability.
- Implemented hybrid retrieval combining BM25 and vector semantic search with legal-term and section-based re-ranking, plus Neo4j-based graph retrieval, with result fusion and deduplication (reduced 10 duplicate results to 1 unique result in testing).
- Applied Act/section metadata filtering to constrain retrieval to the correct legal source, and used RAGAS-based retrieval evaluation plus DeepEval Faithfulness scoring for hallucination detection, raising RAGAS Context Precision from 0.63 to 0.78 and Recall from 0.90 to 1.00 across a validation query set.
- Used LangGraph to orchestrate query analysis, retrieval, graph search, ranking, and generation, with guardrails for prompt injection and jailbreak detection blocking malicious queries before retrieval (0 documents retrieved on blocked requests).
- Achieved ~3.85s end-to-end response latency in observability testing; validated with 3/3 regression tests passing after codebase cleanup.

- Exposed the application through FastAPI, built a Streamlit interface, containerized it with Docker, and deployed it on AWS EC2.

Akki Fashion Assistant — GenAI Multi-Agent Recommendation System

Python, LangGraph, LangChain, OpenAI, ChromaDB, BM25, FastAPI, Streamlit, Docker, AWS EC2

- Built a GenAI-powered product recommendation system for product discovery, comparison, and deal identification.
- Developed a ReAct-based multi-agent workflow with LangGraph covering product discovery, retrieval, ranking, and recommendation generation.
- Combined ChromaDB vector search with BM25 keyword retrieval; exposed the system via FastAPI and Streamlit, containerized with Docker, and deployed on AWS EC2.

Government Scholarship Assistant — RAG-Based GenAI Application

Python, LangChain, ChromaDB, FastAPI, Streamlit, Docker, AWS EC2, Pandas

- Built a RAG-based assistant for discovering government scholarship schemes across NSP, Uttar Pradesh, Delhi, and Uttarakhand.
- Combined structured Excel data with unstructured PDF documents into a unified searchable ChromaDB knowledge base.
- Implemented document chunking and MMR retrieval to handle eligibility queries across overlapping scholarship schemes; deployed via FastAPI, Streamlit, and Docker on AWS EC2.

SmartLoan AI — ML-Based Loan Approval Prediction

Python, Scikit-learn, XGBoost, Pandas, NumPy, MLflow, FastAPI, Docker, GitHub Actions, AWS EC2

- Built an end-to-end loan approval prediction pipeline comparing Logistic Regression, Random Forest, and XGBoost models.
- Used MLflow for experiment tracking and model comparison; served predictions through a containerized FastAPI service.
- Automated build and deployment using GitHub Actions and AWS EC2.

PROFESSIONAL EXPERIENCE

SME Process Specialist — iEnergizer

Jan 2023 – Feb 2025

- Conducted quality audits on 4–5 escalation calls daily and delivered structured coaching focused on call quality and first-call resolution.
- Managed and mentored 20+ agents through training, process coaching, and pre-shift briefings.
- Monitored real-time queues and resolved operational bottlenecks to maintain SLA compliance.

Senior Executive – Voice Support — HCL Technologies BPO Services

Oct 2020 – Feb 2022

- Provided technical and customer support for Pearson UK and US operations, covering examination scheduling, payment processing, and technical issue resolution.
- Worked against AHT, CSAT, and FCR targets while resolving technical issues during online examinations.

Customer Care Executive – Visa Process — iEnergizer

Feb 2019 – Apr 2020

- Provided voice, email, and chat support for visa-related processes across multiple geographies.

Representative, Operations – Telecom Support — Concentrix Daksh Services India Pvt. Ltd.

May 2018 – Feb 2019

- Provided inbound technical support for internet, account, and payment-related issues while meeting customer satisfaction targets.

CERTIFICATIONS

- Oracle Cloud Infrastructure Certified AI Foundations Associate — Oracle University, 2026
- Microsoft Certified: Azure AI Fundamentals (AI-900)
- IBM Data Science Professional Certificate — IBM / Coursera
- Generative AI Mastermind — Outskill

EDUCATION

Bachelor of Commerce

Maharaja Agrasen Garhwal University | 2021 – 2024 (Completed)

Bachelor of Computer Applications — Data Science Specialization

Amity University Online | 2024 – Expected 2028 (In Progress)